

Changing climate states and stability: from Pliocene to present

V. N. Livina · F. Kwasniok · G. Lohmann ·
J. W. Kantelhardt · T. M. Lenton

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Abstract We present a recently developed method of potential analysis of time series data, which comprises (1) derivation of the number of distinct global states of a system from time series data, and (2) derivation of the potential coefficients describing the location and stability of these states, using the unscented Kalman filter (UKF). We test the method on artificial data and then apply it to climate records spanning progressively shorter time periods from 5.3 Myr ago to the recent observational record. We detect various changes in the number and stability of states in the climate system. The onset of Northern Hemisphere glaciation roughly 3 Myr BP is detected as the appearance of a second climate state. During the last ice age in Greenland, there is a bifurcation representing the loss of stability of the warm interstadial state, followed by the total loss of this state around 25 kyr BP. The Holocene is generally characterized by a single stable climate state, especially at large scales. However, in the historical record, at the regional scale, the European monthly temperature anomaly temporarily exhibits a second, highly degenerate

(unstable) state during the latter half of the eighteenth century. At the global scale, temperature is currently undergoing a forced movement of a single stable state rather than a bifurcation. The method can be applied to a wide range of geophysical systems with time series of sufficient length and temporal resolution, to look for bifurcations and their precursors.

Keywords Potential analysis · Bifurcations · Time series analysis · Climate states

1 Introduction

In the past, the Earth's climate has undergone some pronounced changes at both regional and (less frequently) global scales, sometimes in response to relatively small changes in forcing, and sometimes due entirely to internal dynamics. Now human activities are changing the climate and there is concern that, in concert with natural variability, we may have the power to push components of the Earth system into qualitatively different modes of operation, implying large-scale impacts on human and ecological systems (Lenton et al. 2008). Key scientific challenges are to identify and correctly characterize such changes in the past, and to anticipate them in the future.

Pronounced climate changes can come about in at least three ways (Fig. 1): First there are changes that represent the movement of a stable state due to a change in forcing, which may be amplified by constrained positive feedback. Secondly, there are switches between pre-existing, alternative stable states caused by noise (internal variability) in the climate system, which do not require any change in forcing boundary conditions. For these alternative stable states to co-exist under the same forcing, there must be a

V. N. Livina (✉) · T. M. Lenton
School of Environmental Sciences, University of East Anglia,
Norwich NR4 7TJ, UK
e-mail: v.livina@uea.ac.uk

F. Kwasniok
College of Engineering, Mathematics and Physical Sciences,
University of Exeter, Exeter EX4 4QF, UK

G. Lohmann
Alfred Wegener Institute for Polar and Marine Research,
27570 Bremerhaven, Germany

J. W. Kantelhardt
Institute of Physics, Theory group, Martin-Luther-Universität
Halle-Wittenberg, 06099 Halle, Germany

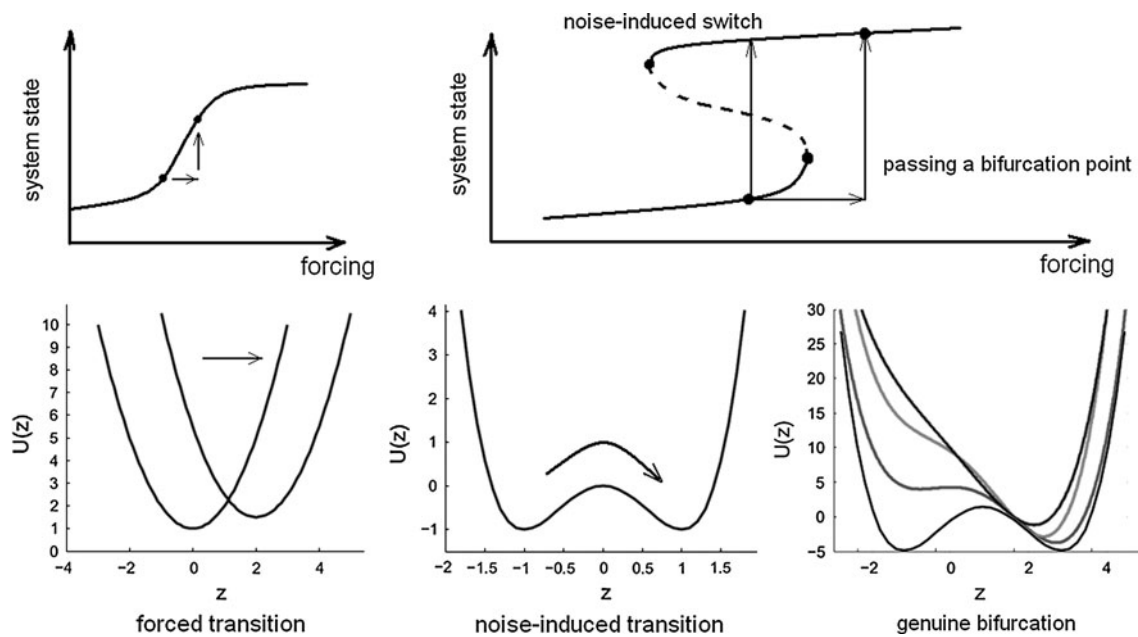


Fig. 1 Schematic of transitions and bifurcations: *upper panels*—bifurcation diagrams describing transitions and bifurcations; *bottom panels*—potential curves

domain of runaway positive feedback in the system. Thirdly, there are changes that involve passing a bifurcation point, in which new climate states become stable or lose their stability.

Here we are particularly interested in detecting changes in the number of states in the climate system and changes in their stability (i.e., bifurcations). Under climate forcing (be it anthropogenic or natural), we expect the shape of the potential that describes alternative climate states to vary, and in some cases the number of climate states to change. We have sought a method that can distinguish bifurcations from noise-induced switches between states, or forcing-induced shifts of a given state. The method we develop is based on detecting the number of states of a system, and their stability or degeneracy, from a given window of timeseries data. As such, it relies on the system in question sampling (i.e. spending some time in) its different states, should they exist, because it is being pushed around by internal variability (noise).

Broadly speaking, our approach can be traced back to the work of (Hasselmann 1976) and others who introduced a conceptual, stochastic approach to climate modelling. Stochastic climate equations and potentials have been studied by means of stochastic resonance (Benzi et al. 1983; Gammaitoni 1998), or general noise-induced perturbations (Ditlevsen 1999; Ditlevsen et al. 2005). Alternatively, (Palmer 1999) characterized the climate system in terms of a general dynamical equation, and showed that it responds to constant, weak external forcing (e.g., anthropogenic increase in radiative forcing) primarily through a

change in the probability density function of the principal components associated with quasi-stationary states. In general, it is crucial to take into account both natural climate variability (Ghil 2002) and external forcing which may influence the stability of the system (Ghil 2000).

In studies of paleoclimate change, it is common to describe Dansgaard-Oeschger (D-O) events during the last ice age as shifts between different states in a stochastically driven, nonlinear system (Alley 2003; Ganopolski and Rahmstorf 2001). Braun et al. (2008), Dima and Lohmann (2008) and Kwasniok and Lohmann (2009) recently introduced conceptual models for Dansgaard-Oeschger events. Depending on the number of states of the diagnostics under study (one-well, two-well or higher order), one can use a polynomial potential of various degrees. Similarly, various forcing functions can be introduced using a polynomial potential as a basis.

The purpose of the paper is to diagnose and compare changing climate states over different time frames during the last 5.3 million years, by means of the recently developed method of potential analysis.

In essence, we describe the climate system as a polynomial potential subject to noise, where the potential may vary in shape due to external forcing. However, our approach could be generalized further to include e.g., the case of periodic forcing in addition to stochastic perturbations.

Our method aims to characterize past transitions in the climate system and to examine whether any components of the climate system are approaching (or may already have

passed) a bifurcation point. As such, it complements methods for anticipating bifurcations based on detecting critical slowing down in time series data, associated with flattening of the potential near a critical point (Held and Kleinen 2004; Livina and Lenton 2007; Scheffer et al. 2009).

The paper is organized as follows: In Sect. 2 we describe the methodology. In Sect. 3 we test the method on artificial data. In Sect. 4 we apply it to various climatic time series. The results are discussed in Sect. 5, with conclusions in Sect. 6.

2 Methodology

2.1 Dynamical framework and polynomial potential

It has been suggested (Benzi et al. 1983; Hasselmann 1999; Kravtsov et al. 2005) that a stochastic equation with double-well potential is an appropriate minimal model for characterizing the Earth's climate under certain boundary conditions. This encapsulates the proposition that the climate system can “oscillate” between two stable states defined by the potential wells (possibly of different levels). It may be particularly appropriate for the D-O events during the last ice age (Ganopolski and Rahmstorf 2002).

Following this approach, Kwasniok and Lohmann (2009) studied the stochastic differential equation

$$\dot{z}(t) = -U'(z) + \sigma\eta, \quad (1)$$

with an asymmetric double-well potential

$$U(z) = a_4 z^4 + a_3 z^3 + a_2 z^2 + a_1 z,$$

and proposed a numerical method to estimate the coefficients of the potential from an observed time series using the unscented Kalman filter (UKF) (see Julier and Uhlmann 2004; Kwasniok and Lohmann 2009; Voss et al. 2004). Assuming a double-well potential model in the NGRIP Greenland ice-core record, they showed that the system has one stable cold stadial mode and one almost degenerate stable warm interstadial mode separated by a potential barrier. The application of this method requires the series to be relatively stationary (i.e., the system does not undergo significant changes or bifurcations), and it requires a prescribed form of the potential (usually a polynomial of 4th degree for a double-well potential with unknown coefficients).

Real systems, however, may be more complex than what a fixed model describes: potentials can oscillate and even change in structure (bifurcate). Therefore, we recently developed a heuristic method (Livina et al. 2010) for detecting the number of states in a system from its time series. Here that approach is used jointly with the method

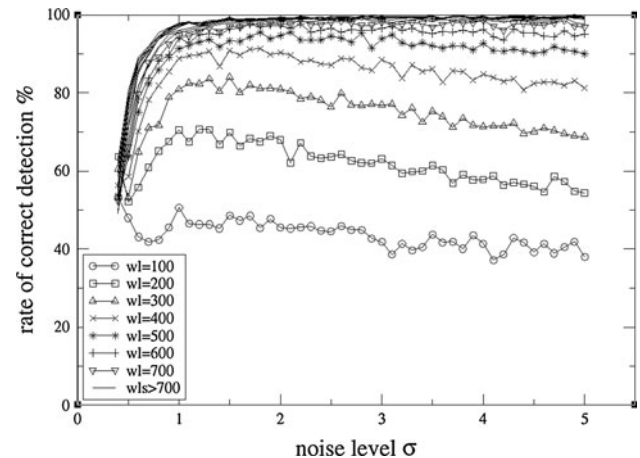


Fig. 2 The rate of detection of two wells in artificial double-well-potential data, $U(z) = z^4 - 2z^2$, with noise level σ values between 0.4 and 5 (step 0.1, 1,000 samples of length 10K for each value of σ). Detection is performed in windows of increasing size, from 100 to 2,000 with step 100. The two potential wells are of equal depths 1, so for small values of σ the potential barrier may be not crossed, and the data in small windows may remain one-well-potential locally—this explains poorer performance of the method at small noise level. For windows bigger than 700 datapoints the correct detection rate is higher than 95% for noise level above 1.5, as one can see that all the curves coincide close to 100% detection. With increasing noise level σ , for small windows the detection rate slightly declines, but still remains higher than 95% for windows bigger than 700 datapoints

of Kwasniok and Lohmann (2009), to allow us to derive the exact potential equation at different stages of the system evolution.

The polynomial fitting of the probability density in sliding windows of varying size provides a quick dynamical portrait of the time series, which is not achievable by mere calculation of potential coefficients over multiple subsets of time series. The potential contour plot is a visualisation of the dynamics (often hidden in time series and not visible by eye) at different time scales. It is possible to run the UKF algorithm for a fixed high-order polynomial, calculating all coefficients and then showing which of them are practically zero and thus estimating the actual structure of the potential. However, to analyse this information, one needs a visualization, and this is where the potential contour plot serves the purpose. Moreover, when the statistics are poor (less than 700–800 datapoints), the polynomial fitting is still capable of detecting the number of states correctly (see Fig. 2). Thus, in this paper we present the full version of the method, combining the dynamical portrait of the series (potential contour plot) and derivation of the potential coefficients (UKF algorithm).

We treat the climate system as a nonlinear dynamical system which can possess multiple states, with shifts between these different climate states induced by stochastic forcing. The state variable z in Eq. 1 represents a large-scale climate variable, for instance, a paleo proxy (or

observational aggregate) measure of temperature over a large region.

The shape of the potential is given by a polynomial

$$U(z) = \sum_{i=1}^L a_i z^i, \quad (2)$$

where the order L is even and the leading coefficient a_L is positive for Eq. 1 to possess a stationary solution. The order of the polynomial controls the complexity of the potential. Increasing values of L allow more states to be accommodated.

2.2 Estimating the number of system states

The number of system states is estimated by means of a polynomial fit of the probability density function of the data. Suppose the system is governed by Eq. 1. The corresponding Fokker-Planck equation for the probability density function $p(z, t)$

$$\partial_t p(z, t) = \partial_z [U'(z)p(z, t)] + \frac{1}{2} \sigma^2 \partial_z^2 p(z, t) \quad (3)$$

has a stationary solution given by (see Gardiner 2004)

$$p(z) \sim \exp[-2U(z)/\sigma^2]. \quad (4)$$

Given this one-to-one correspondence between the potential and the stationary probability density of the system, the potential can be reconstructed from time series data of the system as

$$U(z) = -\frac{\sigma^2}{2} \log p_d(z), \quad (5)$$

where p_d is the empirical probability density of the data. This suggests a least-square fit of $-\log p_d$ to determine the shape of the potential. In order to find the appropriate polynomial degree and thus the number of states in the system, polynomials of increasing degree are fitted and the highest degree before first obtaining a negative leading coefficient is adopted.

This simple model is an approximation of the data, and in reality the dynamics may be far more complex than a potential with added white noise (for example, the noise may be multiplicative and/or coloured). By applying our method we achieve a simple decomposition of the signal into potential and noise component and estimate the modality of the probability density. The method serves practical purposes of identifying potentiality of data, as we recently demonstrated in a set of “blind tests” (Livina et al. submitted).

The number of states present in the system is determined as follows. The empirical probability density p_d of the data is estimated using a standard Gaussian kernel estimator (Silverman 1986). Then least-square fits of $-\log p_d$

(weighted with the probability density of the data (Kwasniok and Lohmann 2009)) with polynomials of increasing even order L are calculated, starting with $L = 2$, until a negative leading coefficient a_L is encountered. The polynomial of highest degree before first obtaining a negative leading coefficient is considered the most appropriate representation of the probability density of the time series, both locally and globally, avoiding overfitting of sampling fluctuations in the probability density. The number of states S in the system is then determined as

$$S = 1 + \frac{I}{2}, \quad (6)$$

where I is the number of inflection points of the fitted polynomial potential of appropriate degree L as described above.

This definition takes into account not only the degree of the polynomial but its actual shape, and we only look at even-order potentials with a positive leading coefficient. These have positive curvature both at minus and plus infinity. Thus, inflection points can only occur in pairs (if any). Any potential has at least one state (with no inflection points). Then we count one further state for each pair of inflection points. This can be either a real minimum (well) or just a flattening in the potential (corresponding to degeneracies in the potential); definition (6) systematically accommodates both possibilities. The number of inflection points is numerically given as the number of sign changes in the second derivative.

After the occurrence of a negative leading coefficient, it may happen that a higher-order polynomial again has a positive leading coefficient. However, it appears reasonable not to consider those higher-order polynomials after the first occurrence of a negative leading coefficient.

A simplified version of definition (6) would be to only count the relative minima (real wells) in the potential. We opt here for the more comprehensive definition of system states as degenerate potentials have been shown to occur in the context of ice-core records (Kwasniok and Lohmann 2009, 2010).

To plot the results with the estimated number of system states, we use a special contour plot, where the x -axis is the time scale of the record, the y -axis is the length of the sliding window moving along the record, and the color denotes the number of detected states, mapped into the middle of the sliding window in the x -axis. Here, red colour denotes one detected state, green—two, cyan—three, purple—four. Note that there is an ambiguity in mapping the results (beginning, middle or end of the windows), and since we consider data aggregated for kernel distribution in each window, it is logical to map the result into the middle.

This 2D contour plot can be used to help detect bifurcations in the time series. A change in the number of

states (vertical change of colour) along all or most time scales (window sizes) indicates the appearance or disappearance of a state (or states). This is not precisely equivalent to a bifurcation, which occurs when a marginally stable state becomes degenerate or a degenerate state becomes marginally stable (i.e. the potential exhibits a plateau or point of inflection). Instead, when the colour change indicates a decline in the number of states, then one can infer that a bifurcation occurred sometime beforehand. Conversely, when the colour change indicates an increase in the number of states, then one can infer that a bifurcation may be approaching. Where e.g. forcing is strong, the change in the number of states may be closely associated in time with a bifurcation, but there is no guarantee of this. To characterize the changing stability of states and thus help locate bifurcations, we must estimate the potential shape.

The sliding windows should be larger than the time scale of jumps between the different states, for the system to be likely to have explored the whole phase space (visited all existing states) in that time frame. When the sliding windows are too small, they may detect a smaller number of states than in the global potential, due to the temporal scale restriction.

The described procedure is used to estimate the number of the system states only. To calculate the potential coefficients, we use the UKF, which is an independent method. For instance, consider a degenerate potential, where two wells bifurcate into one; the method estimating the number of states will correctly identify it as one-well-potential, but since the potential curve may still be asymmetric, it may still require four coefficients (some of them equal to zero) to be described accurately.

2.3 Estimating potential coefficients using the unscented Kalman filter

The estimation of the number of system states is augmented by a nonlinear dynamical estimation of the shape of the potential. The numerical method to derive the coefficients of the potential uses the unscented Kalman filter (UKF) (Julier and Uhlmann 2004; Kwasniok and Lohmann 2009, 2010; Sitz et al. 2002). The method incorporates measurement noise and model error or inadequacy.

The UKF is a nonlinear extension of the conventional Kalman filter. It allows for recursive estimation of unobserved states and parameters in both deterministic and stochastic nonlinear models from incomplete, indirect, and noisy observations. Unlike the extended Kalman filter, the UKF does not linearize the system dynamics but keeps its full nonlinearity. It consistently propagates the first and second moments of the state and parameter estimates.

The dynamical evolution of the climate variable is given by a discretization of Eq. 1 using the Euler scheme with step size h :

$$z_t = z_{t-1} - hU'(z_{t-1}) + \sqrt{h}\sigma\eta_t \quad (7)$$

The observation equation is

$$y_t = z_t + \epsilon_t. \quad (8)$$

y_t is the observed variable which is here identified with the climate records; the state variable z_t is not directly observable. The Gaussian observational noise ϵ_t with zero mean and variance R captures measurement uncertainty and/or model error. The UKF simultaneously estimates the unobserved state z_t and the potential parameters $a_1, \dots, a_L$ from only a time series of the noisy observations y_t . An augmented state vector $\mathbf{x}$ of dimension $n = L + 1$ is created by merging the state variable and the parameters: $\mathbf{x} = (z, a_1, \dots, a_L)$. The dynamical evolution of $\mathbf{x}$ is given by Eq. 7 augmented by a constant dynamics for the parameters.

Let $\hat{\mathbf{x}}_{t-1|t-1}$ be the estimate of the augmented state vector at time $t - 1$ having processed all data up to time $t - 1$ and $\mathbf{P}_{t-1|t-1}^{xx}$ its covariance matrix. The probability density of the augmented state vector is represented by $2n$ carefully chosen so-called sigma points (Julier and Uhlmann 2004). The sigma points are propagated through the augmented dynamical equation, leading to transformed means and covariances $\hat{\mathbf{x}}_{t|t-1}$, $\hat{y}_{t|t-1}$, $\mathbf{P}_{t|t-1}^{xx}$, $\mathbf{P}_{t|t-1}^{xy}$ and $\mathbf{P}_{t|t-1}^{yy}$. When reaching a new observation y_t , the estimates of the state and the parameters are updated using the ordinary Kalman update equations

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t(y_t - \hat{y}_{t|t-1}) \quad (9)$$

$$\mathbf{P}_{t|t}^{xx} = \mathbf{P}_{t|t-1}^{xx} - \mathbf{K}_t \mathbf{P}_{t|t-1}^{xy} \mathbf{K}_t^T \quad (10)$$

where $\mathbf{K}_t$ is the Kalman gain matrix given by

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}^{xy} (\mathbf{P}_{t|t-1}^{yy})^{-1}. \quad (11)$$

See (Kwasniok and Lohmann 2009) for more technical details on the estimation of potentials with the UKF.

2.4 Estimating the noise level

Finally, accurate estimation of the noise level is crucial for the determination of the potential coefficients. To this end, we first apply wavelet denoising of the series with soft thresholding and Daubechies wavelets of 4th order. By subtracting this series from the initial record, we obtain a time series that is used for estimation of the dynamical noise level σ . Using this series, the UKF algorithm is applied in a set of runs with a range of noise levels centered at the roughly estimated value of σ . Then for the coefficients obtained in UKF algorithm, the probability

distribution is calculated, and the most appropriate value of the noise level is chosen by comparison of cumulative distributions of models and data Φ_m and Φ_d (the latter is the integral of the empirical probability distribution p_d) at the minimum value of measure

$$D = \max_z |\Phi_m(z) - \Phi_d(z)|. \quad (12)$$

This measure allows one to find the value of the noise level that provides the minimal global discrepancy between modelled and observed cumulative distributions, i.e. to find empirically the most plausible noise level given the time series of the system.

Wavelet denoising separates the signal into two components, the potential and the noise. They are approximated by our simple model but not exactly, because in a real system the noise may not be white, and the potential is generally nonstationary. When the noise is red with high fluctuation exponent, it possesses nonstationary behavior itself, and the suggested denoising incorporates the appearing local trends into the potential component. Thus, wavelet denoising allows us to apply the method in more general situations which are less idealised than that described by the simple model.

The noise is separated only for the purpose of estimating the noise level σ , which is further used for calculation of the potential coefficients. The noise is never separated when we calculate the potential contour plot. Thus, we apply the separation of noise to eliminate the influence of nonstationarities on the estimation of noise level.

2.5 Note on detrending of geophysical series

As was mentioned above, wavelet denoising separates the noise component in the record, and the obtained series is used for the estimation of the noise level for derivation of the potential coefficients. However, the series might have a low-frequency trend as well as underlying potential dynamics, and it is necessary to pre-process data in such cases before deducing the order of the polynomial (i.e. the number of system states).

We suggest that visible low-frequency trends should be eliminated by subtracting a moving average of the series (with appropriate window size). This differs from wavelet denoising, because a moving average is not as precise in approximating series as wavelet decomposition, and it serves only in initial detrending of the data, without affecting the estimation of the number of states. Indeed, if the series has a long-term trend, the estimation of the number of states using a histogram may be affected by the trend, especially in large time windows (histogram aggregates data, and several shifted subsets of bivariate histogram may overlap as univariate). In such cases it could be

useful to apply potential analysis in a fixed window and monitor the dynamics of potential coefficients and potential curves (see Sect. 3).

Therefore, before the number of states is estimated, any obvious (i.e. detected by eye) long-term trend is removed. However, for further calculation of potential coefficients, which relies on the assumption of quasi-stationarity, the initial data is considered. In this case, the noise estimation is to be done using the wavelet denoising.

2.6 Final method overview

Thus, the proposed method of potential analysis consists of the following two stages:

- Estimation of the number of states in the system, which is visualised in a 2D contour plot. If the series has a visible low-frequency trend, it is pre-processed and the trend removed using a moving average technique.
- Derivation of the potential coefficients, which is visualised in a 3D plot of potential curves. For the estimation of the noise level, wavelet denoising is applied. Chunks of quasi-stationary data are considered without detrending.

The method is first tested on artificial series with known simulated dynamics, to check its reliability; then it is applied to paleoclimate, historical and observational records.

3 Tests

3.1 Test of potential analysis on an ensemble of double-well potential artificial data

We consider an ensemble of double-well potential artificial data with prescribed potential $U(z) = z^4 - 2z^2$, which has two symmetric wells of depth 1. We generate 1,000 samples of length 10,000 data points for each value of the noise level σ in the range from 0.4 to 5 with step 0.1. The sampling interval is 0.05 system time units. We considered data in windows of size from 100 to 2,000 data points (step 100). The results are shown in Fig. 2, where various curves corresponding to varying size of the data window show the rate of accurate detection of two wells in the data versus the level of noise σ . This experiment demonstrates that, first, for the noise level much smaller than the well depth, the detection is less successful (which actually means that the data in this small window is not double-well potential but rather one-well potential). Second, for window size of 500 datapoints and larger, the detection rate is very high

Table 1 Summary of datasets analyzed

Time span	Measured variable	Region	Temporal resolution	Datapoints per segment	Source
5320–0 kyr BP	Benthic $\delta^{18}O$	Global composite	1,000 yr	1,000	Lisiecki and Raymo 2005
800–0 kyr BP	Temperature reconstruction	Antarctica	100 yr	700	Jouzel 2007
60–0 kyr BP	$\delta^{18}O$	Greenland	20 yr	500	Steffensen 2008
60–11 kyr BP	Calcium	Greenland	1 yr	10,000	Fuhrer et al. 1993
14–0.1 kyr BP	Metal concentration	Venezuelan coast	2 yr	1,000	Haug et al. 2001
1165–1990	Foraminifera	Southern Caribbean	0.1 yr	1,000	Black et al. 1999
1659–2004	Temperature reconstruction	Europe	0.083 yr	600	Luterbacher et al. 2004
1880–2009	Temperature anomaly	Global composite	0.083 yr	100	Smith et al. 2008

and thus the method is reliable even for small windows of 500–700 points (which is often the case when we deal with geophysical data, observed or reconstructed; see Table 1). Third, even when the noise level is very high (five times as high as the depth of the potential well) and therefore masks the potential dynamics very significantly, the method successfully detects the underlying potential. This demonstrates the robustness of the algorithm.

3.2 Double-well potential artificial data with nonstationarities

We generate double-well potential data ($U(z) = z^4 - 2z^2$) with nonstationarities using trends in the noise level σ and mean value of the series, see Fig. 3. In the first series, the noise level σ is decreasing from 1.5 to 0.1,

$$\begin{aligned}\dot{z}(t) &= -(z^4 - 2z^2)' + \sigma(t)\eta, \\ \sigma(t) &= -0.00007t + 1.50045, \quad t \in [1, 20000].\end{aligned}$$

In the second series, σ remains constant, but a deterministic forcing with a parabolic trend varying from 0 to 2 is added to the dynamical equation, driving the mean value of the time series:

$$\begin{aligned}\dot{z}(t) &= -(z^4 - 2z^2)' + F(t) + \sigma\eta, \quad F(t) = 5 \cdot 10^{-9}t^2, \\ t &\in [1, 20000].\end{aligned}$$

To monitor the dynamics of the potential, we considered a sliding window of 5,000 datapoints length, and in each subset we derived the potential curves and potential coefficients $a_1, \dots, a_4$. Plotting the dynamics of the coefficients allows us to evaluate the influence of the nonstationarities on the shape and analytical expression of the potential, and may be used to estimate what parameters are to be introduced for bifurcation analysis.

The series with decreasing noise level (Fig. 3a) and with fixed noise level and imposed parabolic trend (Fig. 3b) have distinctively different dynamics of the potential sets (Fig. 3c,d). Accordingly, their potential coefficients change differently (Fig. 3e,f): whereas the former series has a_1, a_3 (odd powers of variable)

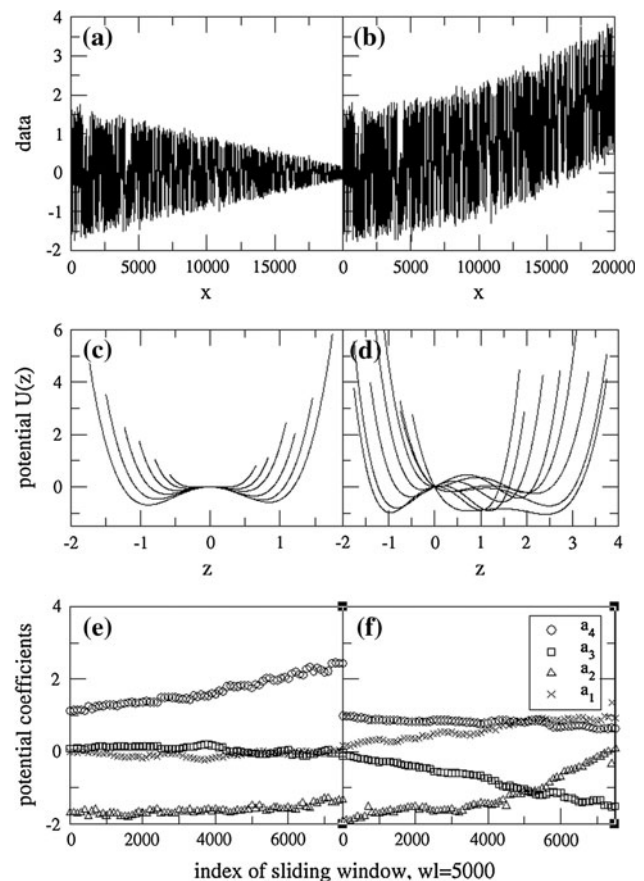


Fig. 3 Two samples of double-well-potential noise with nonstationarities: decreasing noise level and imposed parabolic trend (a, b). The potential curves in c, d are obtained in the subsets of data using a sliding window of 5,000 datapoints length (only a few curves are shown for clarity). The dynamics of corresponding coefficients $a_1, \dots, a_4$ along the series are plotted in (e, f)

effectively close to zero, the latter series has all the four coefficients changing (a_1, a_3 appearing from initially zero values), and the potential dynamics can be described as more complex: not only changing shape, but also moving along the z axis (this is in agreement with the known trend in the data).

These two series demonstrate the difference between a genuine bifurcation (in the first series, double-well potential behaviour tends to one-well potential) and a forced transition (the structure of the potential does not change, but the potential drifts according to the forcing trend). The derived potential coefficients may be used for forecasting the system dynamics.

4 Case studies and results

We consider several climate records of varying temporal resolution and spatial coverage, spanning time intervals from millions of years (Lisiecki and Raymo 2005) benthic $\delta^{18}\text{O}$ stack from 5.3 million years BP) to the recent record of global temperature (monthly data since 1880), in order to search for changes in the number and stability of climate states.

To give a very broad, introductory overview of climate variability over these timescales: Northern Hemisphere glaciations began roughly 3 million years BP and were initially dominated by the 41 kyr variation in the Earth's obliquity. Around 1 million years BP, in what is known as the 'Mid-Brunhes transition', a roughly 100 kyr frequency began to dominate glacial-interglacial variability. During the last of these circa 100 kyr ice ages, we have a detailed record of Northern Hemisphere climate variability and it shows a series of abrupt climate warming events (followed by cooling), known as Dansgaard-Oeschger events. On exiting the last ice age into the Holocene interglacial, the climate became more stable with a lower level of internal variability. However, there have still been some marked climate changes during the Holocene, especially involving the hydrological cycle in parts of the tropics. Over the last millennium, the climate has been stable at the global scale, but at the regional scale there have been some marked shifts, e.g. the 'medieval warm period' and 'little ice age' in Europe. The observational record shows a global increase in temperature, especially over the last 50 years, which is attributed to anthropogenic radiative forcing.

In the cases of uneven or insufficient temporal resolution, we interpolated data (1D linear interpolation using Matlab `interp1` function). The datasets are summarised in Table 1 (the temporal resolution is given after interpolation where applied, see the description of the datasets in the following subsections).

A low-frequency trend is visible in several of the time series (benthic $\delta^{18}\text{O}$ data, Cariaco Basin trace metal data, and NOAA global monthly index). We removed long-term trends in these series using a moving average technique, before estimating the number of system states. In all the case studies, the noise level is estimated using wavelet denoising, and it is then used for derivation of the potential coefficients.

Where a bifurcation is suggested, based on a changing number of detected system states, we show the estimated potential curves for different segments of the time series in question. Where there is no indication of bifurcation, the potential curves are not shown. We reiterate that a detected decline in the number of system states indicates that a bifurcation occurred sometime beforehand. Conversely, a detected increase in the number of system states provides forewarning that a bifurcation may be approaching. This is because the first part of our method (the coloured contour plots) recognizes even highly degenerate (i.e. bifurcated) states.

4.1 The onset of glacial cycles

The benthic $\delta^{18}\text{O}$ stack (Lisiecki and Raymo 2005) (composite sediment core record) spans 5.3 Myr and represents an average of 57 globally distributed records (Fig. 4a). $\delta^{18}\text{O}$ is a proxy for ice volume on the planet and also contains a component due to changes in ocean temperature—increasing $\delta^{18}\text{O}$ indicates increasing ice volume and declining temperature. The composite record is orbitally tuned, so one should be careful in interpreting any apparent periodicity, but this does not affect our analysis here. Since the data has variable resolution (5 kyr in the beginning of the record and 1 kyr at the end), we interpolated it to a common resolution of 1 kyr. As the data is clearly non-stationary, we detrended it using a moving average (window length 100) (Fig. 4b). We calculate potential curves for 5 segments of the data of length 1,000 kyr (starting from 5,000 kyr BP, see Fig. 4e).

Initially, the climate is mostly characterized as having a single state, which is particularly stable over the interval 4,000–3,000 kyr BP as shown by the deep potential well for segment II (Fig. 4e). An increase in the number of climate states from one to two is detected across a wide range of timescales at about 3,000 kyr BP (Fig. 4d), accompanying the start of significant Northern Hemisphere glaciation. This provides a possible forewarning of bifurcation, representing the separation of glacial and interglacial climate states. However, the potential curve spanning 3,000–2,000 kyr BP (segment III) shows that the states are quite degenerate and indistinct (Fig. 4e). Moving forward to the intervals 2,000–1,000 kyr BP (IV) and 1,000–0 kyr BP (V), the separation of glacial and interglacial states increases, with the glacial state clearly stable, and the interglacial state more indistinct and degenerate. The widely discussed 'Mid-Brunhes transition' at around 900 kyr BP, where the 100 kyr glacial-interglacial cycles first become prominent in the record, does not show up very clearly (Fig. 4d) but perhaps this is to be expected as the method is not designed to detect changes in the frequency domain.

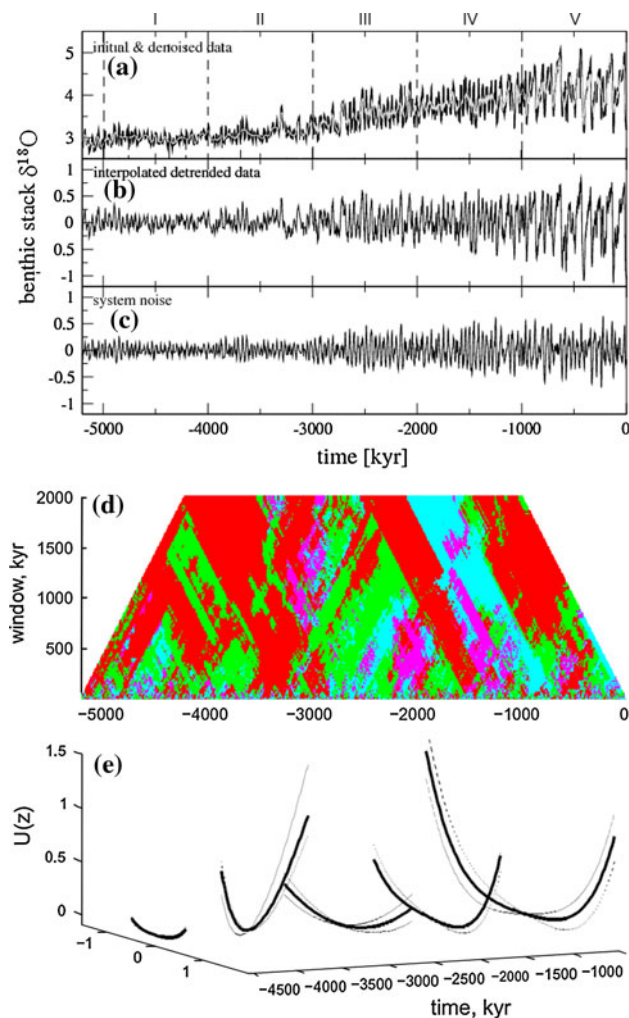


Fig. 4 Raymo benthic data: **a** raw (black) and denoised (grey); **b** detrended data, moving average is removed (window size 100), **c** system noise obtained after wavelet denoising; **d**) 2D potential contour plot of the detrended series; **e**) potential curves of 10 chunks of data denoted by Roman numbers above the series. In this and all other plots of the potential analysis colors indicate the number of detected states as follows: *red*—one well, *green*—two wells, *cyan*—three wells, *purple*—four wells

Considering data for the last 1000 kyr, the red colour in the upper part of the plot (indicating a single state) corresponds to very large windows (1,000–2,000 kyr), which aggregate a lot of data with nonstationarities from previous time periods. This should be distinguished from real bifurcations, where the same number of states are detected at all time scales, including smaller windows (as at 3,000 kyr BP). In the last 1,000 kyr, the smaller time scales have patchy detection, in most cases showing 2 states, which is what one would expect from data spanning glacial and interglacial climates.

It is interesting to note that the noise level, representing internal variability in the climate system, increases with the

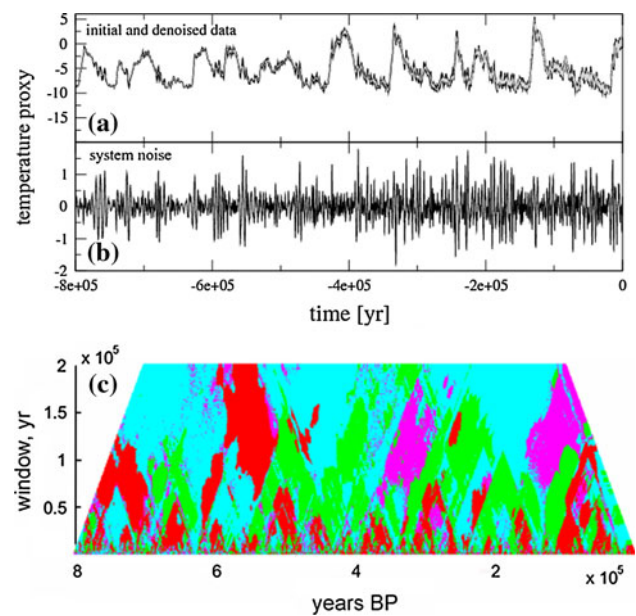


Fig. 5 EPICA data: **a** initial and denoised data; **b** system noise obtained after wavelet denoising; **c** potential analysis of the initial data. The record is highly nonstationary; so it is not surprising that the potential analysis indicates high variability and no homogeneous regime

onset of 41 kyr cycles around 3,000 kyr BP and its maximum amplitude gets larger toward the present. However, the noise level fluctuates considerably, complicating the analysis.

4.2 The 100 kyr world

The EPICA (European Project for Ice Coring in Antarctica) Dome Concordia drilling site is located on the East Antarctic plateau, and the recovered record (Jouzel 2007) spans the time interval from 800 kyr BP (Fig. 5). The temperature reconstruction shown is from the δD proxy. It is unevenly spaced and highly non-stationary but without a clear overall trend; so we consider interpolated data towards resolution 100 yr, but without detrending. Applying wavelet denoising, one can see that the noise level is non-stationary as well, and of small amplitude compared with the data (Fig. 5b).

In fact, this record may not be well captured by a potential function, i.e. a differentiable function with added stochastic component—note in particular the abrupt spikes in the data. Instead, a non-potential function with damping may be more appropriate. This is confirmed by the patchy pattern of estimated number of states, with intermittent 1- or 2-well-potential behavior at shorter time scales and 3- or 4-well at longer time scales (Fig. 5c). This reflects the variability in the data. There is no clear emergence of, for example, interglacial, early/weak glacial, and late/full

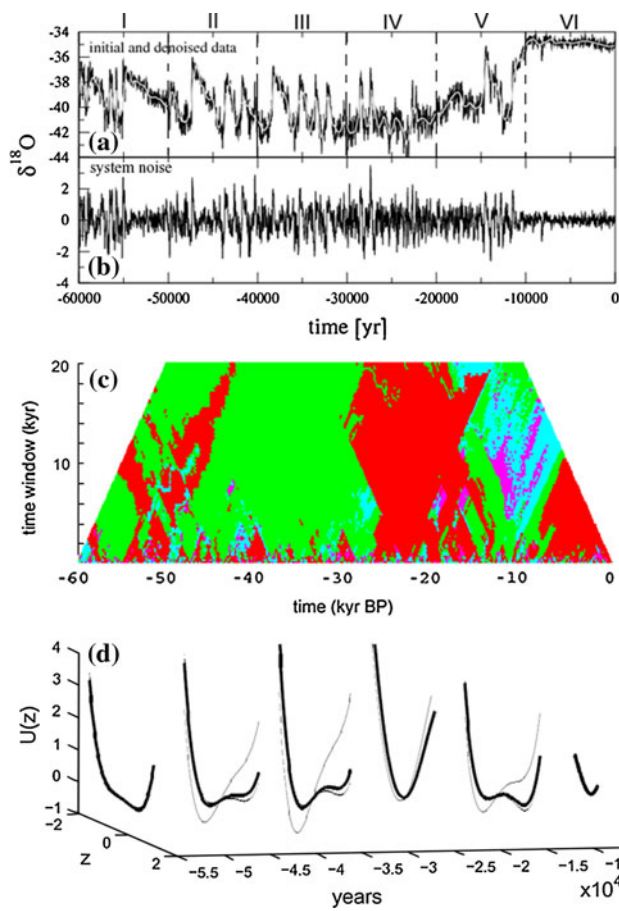


Fig. 6 GRIP ice-core data: **a** raw (black) and denoised (grey); **b** system noise obtained after wavelet denoising; **c** 2D potential contour plot of the initial series; **d** potential curves of 6 chunks of data denoted by Roman numbers above the series

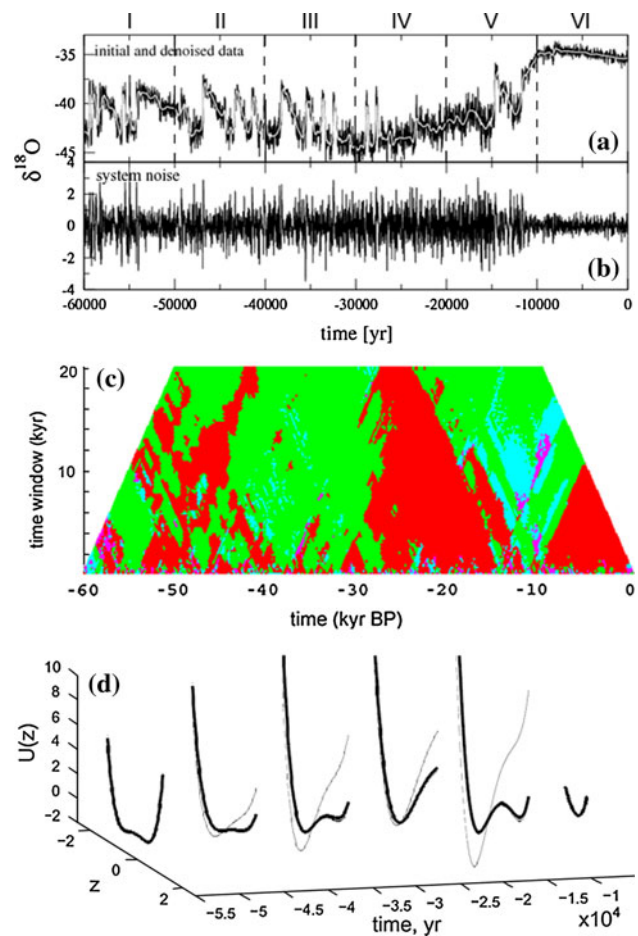


Fig. 7 NGRIP ice-core data: **a** raw (black) and denoised (grey); **b** system noise obtained after wavelet denoising; **c** 2D potential contour plot of the initial series; **d** potential curves of 6 chunks of data denoted by Roman numbers above the series

glacial states as assumed in some models. As recent “blind tests” have shown, potential and non-potential data can be accurately distinguished by our method (Livina et al. [submitted](#)).

4.3 The Dansgaard-Oeschger events

GRIP and NGRIP are Greenland and North Greenland Ice Core projects (Steffensen [2008](#); Niels Bohr) (NGRIP being drilled about 325 km north of the summit core GRIP). We consider ice-core records of stable water isotope $\delta^{18}\text{O}$ from both sites, which is a proxy for temperature (in this case more negative is colder), and spans the time period from 60 kyr BP with temporal resolution 20 yr (Figs. [6a](#), [7a](#)). For comparison, we also consider calcium data from the GRIP project (Fig. [8a](#)), which has annual resolution but ends at 11 kyr BP, and is known to be anti-correlated with the temperature proxy (Ditlevsen [1999](#)). In each case the data are denoised but not detrended.

The data are highly variable and non-stationary, but as in previous work (Livina et al. [2010](#)), we detect the loss of a second climate state at about 25 kyr BP in all three datasets (Figs. [6c](#), [7c](#), [8c](#)). This represents the warm interstadial state becoming degenerate and ultimately being completely lost as ice builds up to the last glacial maximum. We infer that a bifurcation occurred beforehand (because the method still detects a degenerate state as a separate climate state). The bifurcation at (or prior to) 25 kyr BP is detected across all time scales (windows sizes from 50 to 1,000 datapoints) – see the test of the ensemble of artificial data in Fig. [2](#), where the influence of the window size is investigated. The calcium data has annual resolution, so the same time scale corresponds to window sizes from 1,000 to 20,000 datapoints. The detection in three datasets, across a range of timescales, suggests the result is robust.

A further remark on the statistical significance of the bifurcation detection and the required minimum window

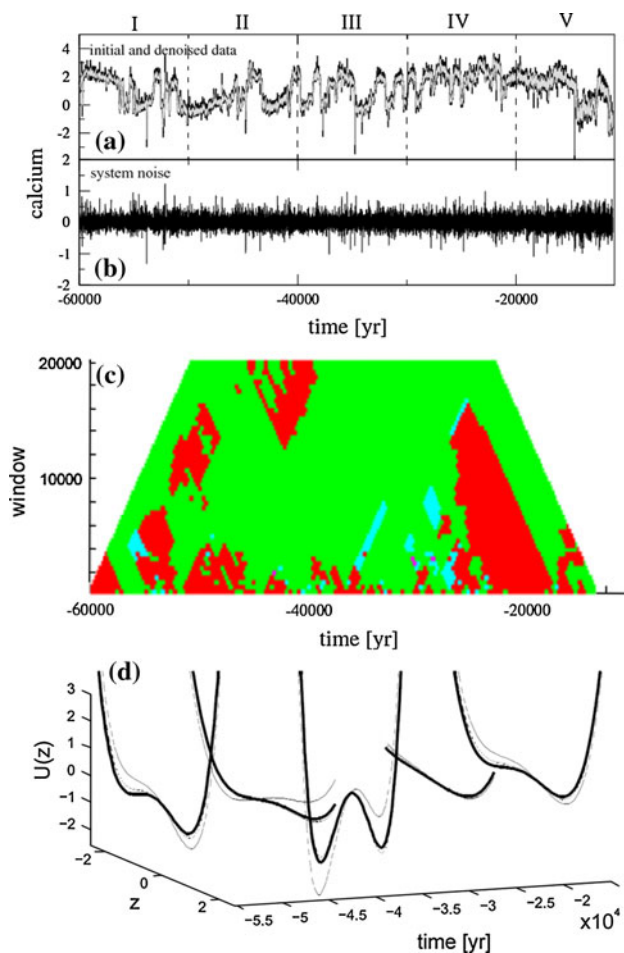


Fig. 8 GRIP calcium data: **a** raw (black) and denoised (grey); **b** system noise obtained after wavelet denoising; **c** 2D potential contour plot of the initial series; **d** potential curves of 5 chunks of data denoted by Roman numbers above the series. The major bifurcation at 25 kyr BP is detected similarly to GRIP and NGRIP $\delta^{18}\text{O}$ records. Although this record starts and ends earlier than those two records, and it has annual resolution (whereas those two have temporal resolution 20 years), we consider the calcium record between 60 and 11 kyr, and with the same sliding windows to make them comparable in time scales

length is in order. As mentioned before, the system needs to sufficiently explore state space within the time window for a correct detection of the number of states present. The waiting times between successive Dansgaard-Oeschger events follow an exponential distribution with a mean waiting time of about 2.7 kyr (Ditlevsen et al. 2005). The probability that the system in any 10 kyr by chance just resides in the cold state, causing an erroneous detection of only one state instead of two, is of order $\exp(-10/2.7) \sim 0.025$. Thus, a window size of 10 kyr is large enough and the bifurcation is significant at a high significance level.

The potential curves for $\delta^{18}\text{O}$ (Figs. 6d, 7d) provide more detail and show that initially 60–50 kyr BP (I), the cold and warm states have comparable stability, but as time progresses, the cold state becomes more stable and the

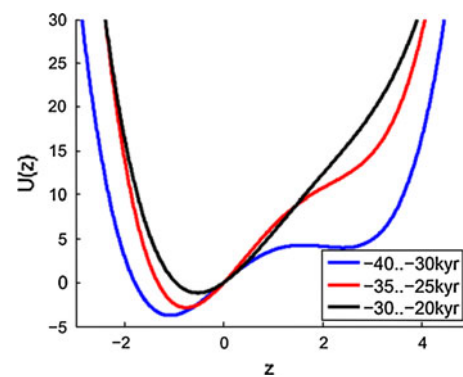


Fig. 9 NGRIP potential curves demonstrating the bifurcation at about 25 kyr BP

warm state less stable. In the interval 30–20 kyr BP (IV), the warm state has disappeared (or is highly degenerate). To examine this a little closer, potential curves for different windows of the NGRIP data are plotted (Fig. 9), which show that the interval 40–30 kyr BP is characterized by a separate, marginally stable warm state, which by 35–25 kyr BP has become degenerate. Looking back at the time series it is apparent that this corresponds to the Dansgaard-Oeschger events becoming shorter in duration.

Examining the dynamics of the extremal points of the NGRIP potential curves for fixed sliding window of 10 kyr (Fig. 10) shows only those bifurcations that produce clearly defined system states: the extremal points can be calculated only when the potential well is deep enough to distinguish between local maxima and minima. Thus, degenerated and marginally-stable states that are detected by the potential contour plot (Fig. 7c) cannot be identified in this plot (Fig. 10). However, when the potential wells become well defined, one can see a change of the number of extremal points from one (one-well potential) to three (double-well potential with one local maximum and two local minima). This reveals that from soon after 60 kyr BP to around 15 kyr BP there is only one strongly stable state, the other state is only marginally stable, in agreement with earlier work (Kwasniok and Lohmann 2009). However, through the deglaciation two stable states (and one unstable one) can be clearly distinguished.

The noise level declines markedly on entering the Holocene, roughly 11 kyr BP (Figs. 6b, 7b), which is characterized by a single, stable climate state (Figs. 6c,d, 7c,d, 10).

4.4 The holocene

Moving forward in time to look in more detail at the Holocene, we consider metal concentration data (Ti, Fe) (Haug et al. 2001) from the sediments of the anoxic Cariaco Basin off the Venezuelan coast at subdecadal resolution.

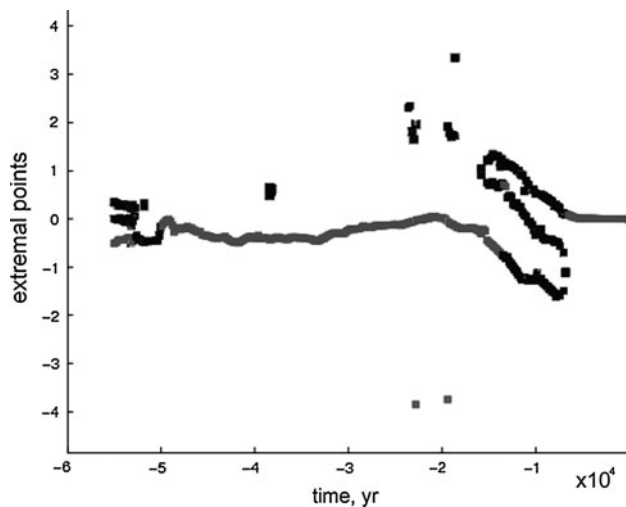


Fig. 10 Extremal points (maxima and minima) of the potential curves of NGRIP data, mapped to the middles of the sliding windows of size 10 kyr. When the potential is single-well, there is only one extremal point (global minimum). When the potential is double-well, there are three extremal points (two minima and one maximum between them). The change of the number of extremal points (from one to three or from three to one) signifies a bifurcation. Note that minima/maxima appear when the bifurcation has passed, because they cannot be calculated when the potential undergoes degeneration (hyperbolic point). Moreover, the number of extremal points cannot be calculated correctly for highly degenerated potential, when a well is very shallow or disappearing. Therefore, the change of the number of states or the moment when the bifurcation occurs cannot be concluded from the plot of extremal points

This record may be considered a proxy of the variations in the hydrological cycle in northern South America corresponding to shifts in the mean latitude of the Atlantic Intertropical Convergence Zone over the last 14 kyr (Fig. 11a). The data clearly has a long-term parabolic trend, which we remove by detrending using a moving average technique with sliding window of length 3 points. After detrending (Fig. 11b), the data show greater variability prior to 11 kyr BP and again in the interval 3.5–2.5 kyr BP, and these are also times of higher noise level (Fig. 11c).

The potential analysis generally shows one stable state (Fig. 11d), in agreement with the relative stability of the Holocene climate and our analysis of the Greenland ice cores (Figs. 6c,d, 7c,d). However, there are hints of the possible appearance of additional states at times of fluctuation in the detrended data, for example, a triple-well potential around 3.5 kyr BP.

4.5 The last millennium

Moving toward the present, the abundance (units of no. of individuals per gram of sediment) of the planktic foraminifera *Globigerina bulloides* in the Southern Caribbean (Black et al. 1999) provides a record of ocean-atmosphere

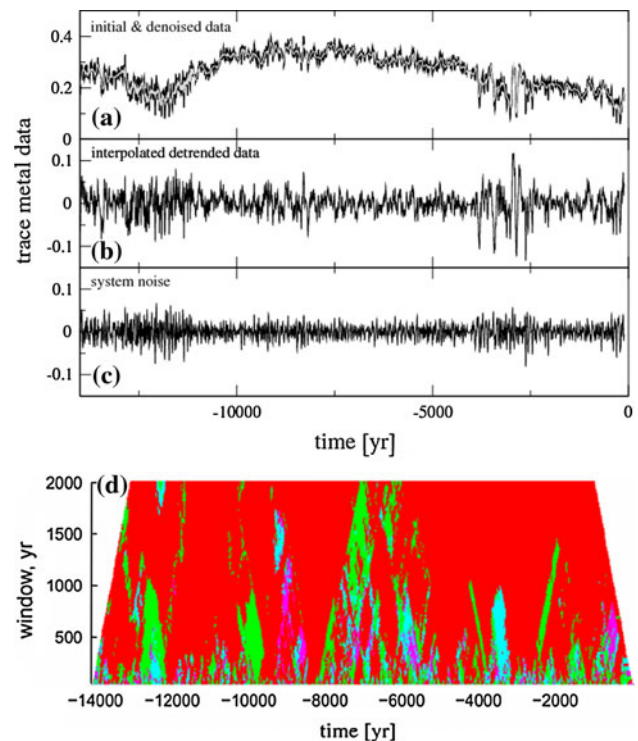


Fig. 11 Cariaco Basin trace metal data: **a** raw (black) and denoised (grey); **b** detrended data, moving average is removed (window size 3), **c** system noise obtained after wavelet denoising; **d** 2D potential contour plot of the detrended series. Potential curves are not shown, because 2D contour plot shows generally one-well-potential behavior with minor oscillations at short time scales. No major bifurcation is observed, which confirms relative stability of the last 10 kyr climate

variability in the North Atlantic basin from 1165 until 1990 (Fig. 12a). This is a proxy of the changes in trade-wind-induced upwelling intensity which corresponds to decadal-scale variability in tropical Atlantic surface conditions. The record exhibits strong decadal to centennial climate variability and part of the variability can be attributed to the Atlantic Multi-decadal Oscillation (AMO) (Lohmann et al. 2004). The data is sparse, comprising 565 points, so we interpolate it to obtain 8,248 points (this particular choice was dictated by the time interval the series spans).

The potential analysis (Fig. 12c) reveals the dominance of a single climate state. One can see a hint of a second state perhaps appearing in the seventeenth century, but the general pattern of the data is one-well-potential.

4.6 Historical European climate anomalies

The averaged European monthly temperature anomaly 1659–2004 (expressed relative to the 1901 to 1995 calibration average) is a reconstruction based on a large number of homogenized and quality-checked instrumental data series, a number of reconstructed sea-ice and

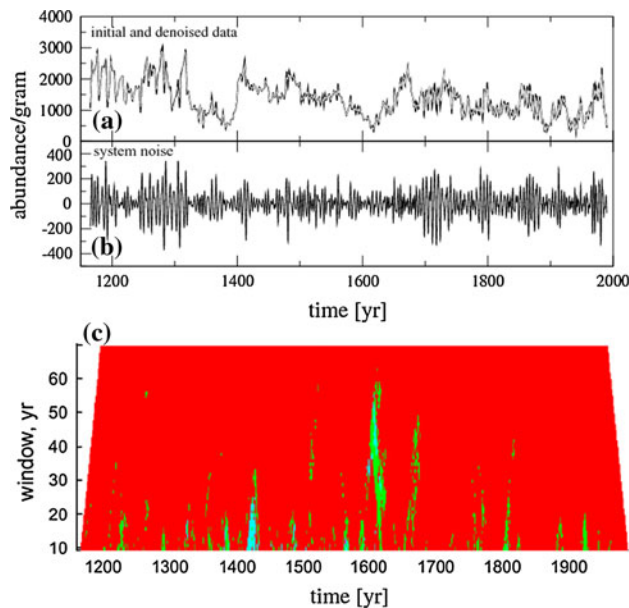


Fig. 12 Cariaco Basin *G. bulloides* abundance data: **a** raw (black) and denoised (grey); **b** system noise obtained after wavelet denoising; **c** 2D potential contour plot of the initial series. Potential curves are not shown, because 2D contour plot shows generally one-well-potential behavior with minor oscillations at short time scales. No major bifurcation is observed except a short-term change for double-well potential behavior in the XVII century

temperature indices derived from documentary records for earlier centuries, and a few seasonally resolved proxy temperature reconstructions from Greenland ice cores and tree rings from Scandinavia and Siberia (Luterbacher et al. 2004) (Fig. 13a). The time series is an average over the land area from 25°W to 40°E and 35°N to 70°N. The values for the period to 1900 are reconstructions; data from 1901 to 1998 are derived from (New et al. 1901). The post-1998 data is from (Hansen 2001).

The potential analysis generally shows a single stable state, but it detects the appearance of a second climate state around 1760 that persists until roughly 1800 (Fig. 13c). The shape of the potential curve for 1750 to 1800 (II) indicates that the predominant climate state became broader and less stable (Fig. 13d), but that the second, colder climate state inferred was highly degenerate. In the first half of the 19th century (III) the climate reverted to a more stable single state. This can also be seen in the histograms of the data for the two intervals, Fig. 14.

4.7 Global temperature

The NOAA global monthly temperature index (Fig. 15a) is a combined land and ocean record, where anomalies are provided as departures from the twentieth century average (1901–2000) (Smith et al. 2008). The series has a well-known, visible trend, which we remove using moving-average

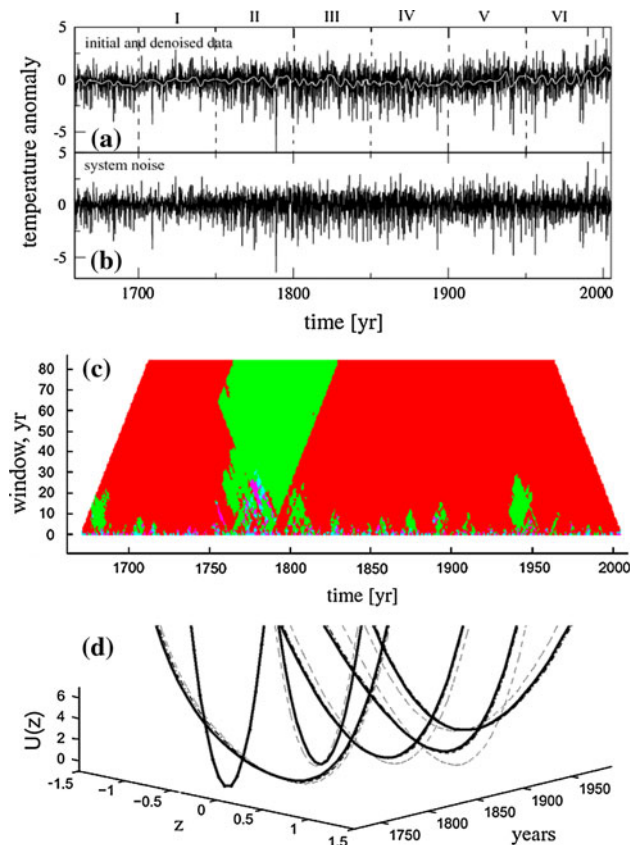


Fig. 13 European monthly temperature anomaly data: **a** raw (black) and denoised (grey); **b** system noise obtained after wavelet denoising; **c** 2D potential contour plot of the initial series; **d** potential curves of 6 chunks of data denoted by Roman numbers above the series. 2D potential contour plot shows bifurcating behavior at the end of the eighteenth century, and this can be seen in the potential curves, where 2nd chunk has highly degenerated 2-well potential shape. This effect is probably explained by the Malda anomaly in the eighteenth century

detrending with window length 10. Interestingly, in the case of this series, the detrended data is almost identical to the series obtained using wavelet denoising, which may indicate the dominating influence of the noise masking the potential.

Our method robustly detects only one climate state, with very minor oscillations at small time scales (Fig. 15d). This indicates that global temperature change is not currently undergoing a bifurcation.

5 Discussion

We have proposed a method of potential analysis of geophysical time series that allows one to estimate the number of states in a time series following the assumption that the dynamics are governed by a stochastic model with a potential function and one noise component. We have tested the method on artificial data and then applied it to several paleo- and historic climatic records to search for

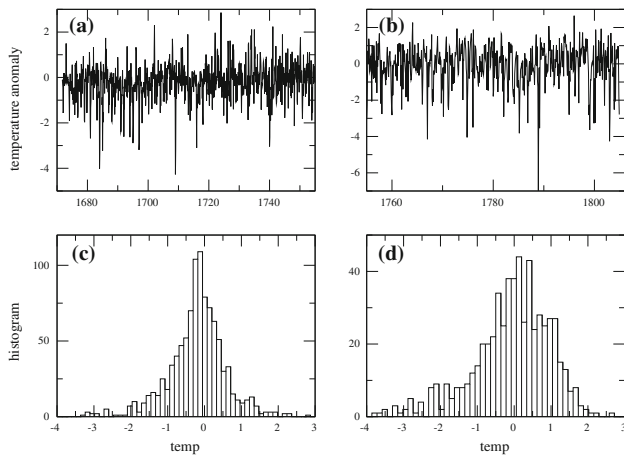


Fig. 14 Two subsets of European monthly temperature anomaly data: **a** time period before appearing second state; **b** time period when the second state appears; **c** histogram of data in **(a)**; **d** histogram for data in **(b)**, note the appearing second state for negative values of temperature anomaly

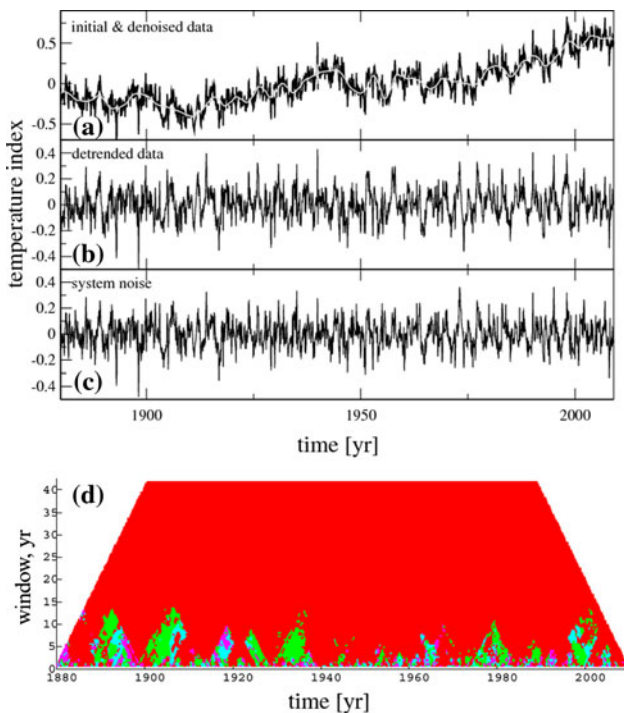


Fig. 15 NOAA global monthly temperature index: **a** raw (black) and denoised (grey); **b** detrended data, moving average is removed (window size 10); **c** system noise obtained after wavelet denoising; **d** 2D potential contour plot of the detrended series

possible bifurcations and distinguish them from noise-induced transitions.

The first occurrence of continental-scale Northern Hemisphere ice sheets, especially on Greenland, is recorded as ice-rafted detritus released from drifting icebergs

into sediments of the mid- and high-latitude ocean. After a transient precursor event at 3.2 Ma, signals of large-scale glaciations started in the subpolar North Atlantic in two steps, at 2.9 and 2.7 Ma, e.g. (Bartoli et al. 2005). The shift at around 3 Ma has been seen in wavelet time series analysis (Lohmann 2009), using a significance test against a red noise background spectrum (Maraun and Kurths 2004). Our new method detects the onset of Northern Hemisphere glaciation as the appearance of a second climate state around 3 Ma. This detection of a new climate state can be observed across all time scales (Fig. 4d), albeit not particularly cleanly. The shape of the corresponding potential for 3 to 2 Ma shows that the two climate states are indistinct (interval III in Fig. 4e). This suggests that the glacial and interglacial stages of 41 kyr obliquity-paced cycles were initially not particularly distinct (the climate spent plenty of time between them).

Subsequently, with ice continuing to build up on the planet, the cold and warm states become more separated (Fig. 4e), with the cold state stable and the warm state degenerate (unstable). The widely recognized Mid Brunhes transition from 41 kyr to circa 100 kyr glacial cycles around 0.9 Ma only shows up as a broadening of the potential (contrast IV and V in Fig. 4e) with no clear change in the number of climate states (Fig. 4d). The error range on the reconstructed potential for the last 1 Myr (V in Fig. 4e), indicates that either the cold (glacial) or warm (interglacial) state could be characterized as stable. This may reflect the influence of long interglacial periods such as Marine Isotope Stage 11 around 420 kyr BP.

Analysis of the EPICA Antarctic temperature proxy record from 800 kyr BP gives an unclear picture. Across the longest time windows, the predominant result is a system with three states. For the past 400 kyr or so these states might be identified as interglacial, full glacial and early glacial, as in the model of Paillard (1998). However, the early part of the record, when there is nothing corresponding to the current warm interglacials, is also seen as three states, and temporarily (circa 600 to 500 kyr BP) as one state. We conclude that this record is not well described by a potential function.

Looking at climate variability within the last ice age, as recorded in Greenland, a clearer picture emerges: Two distinct climate states are detected from 60 kyr BP to around 25 kyr BP, corresponding to the cold stadial and warm interstadial. They start out with similar stability, but the cold stadial state takes over dominance, and then the interstadial state becomes degenerate and ultimately disappears. The reduction to one state is detected clearly around 25 kyr BP in both GRIP and NGRIP oxygen isotope records and in GRIP calcium data. The bifurcation point at which the warm interstadial state becomes degenerate occurs before this, but pinning it down is

difficult. The potential generated from data spanning 40 to 30 kyr BP gives a marginally stable interstadial state, whereas for data spanning 35 to 25 kyr BP the interstadial state is degenerate, suggesting bifurcation occurs in the interval 30 to 25 kyr BP and certainly after 35 kyr BP. Two stable climate states clearly reappear through the last deglaciation, before one disappears again (Fig. 10).

Our analysis of the Greenland ice core records supports the view that the last circa 10 kyr of the Holocene has a single stable climate state in the northern high latitudes. Analysis of a tropical, regional record of hydrological cycle variability over northern South America, also supports a single Holocene climate state as the dominant result. However, there are intervals, e.g. around 3.5 kyr BP, where multiple states are detected over a time window of up to 1,000 years. Much recent research has highlighted variability in the hydrological cycle in the tropics during the Holocene, as an agent in the rise and fall of ancient civilisations. In this particular regional case, the climate change around 3.5 kyr BP may be linked to the rise of an Andean civilization (Binford et al. 1997).

In the historical record, analysis of European monthly temperature anomalies suggests a second, degenerate climate state began to appear during the interval roughly 1760–1800 (Fig. 13c). At this time, the European climate was recovering from the Little Ice Age—the last ice fair on the Thames took place in 1816. Striking regional climate anomalies in the interval 1760–1800 have previously been recognized and named the “Malda anomaly” after Baron de Malda who recorded observational notes (Barriendos and Llasat 2003). The Mediterranean climate was particularly anomalous at this time, with thunderstorms, floods, droughts, and severe winters. However, the method may simply be picking up an interval during which a series of large volcanic eruptions generated anomalous cold events.

Moving forward in time and outward in scale, to the observational record of global temperature, there is no sign of multiple climate states or bifurcation. Instead the current global temperature rise is best interpreted as movement of a single stable state under anthropogenic (plus natural) forcing mechanisms. However, more careful examination of regional scale records for the appearance of new climate states is warranted.

Our method of potential analysis carries several caveats. We have opted for a minimal model of a polynomial potential with stochastic noise, but this may be inappropriate for some climate systems and their time series. The clearest example herein are the recent circa 100 kyr glacial cycles as recorded in the EPICA ice core, where a potential model appears inappropriate. For other systems, additions to the model may be appropriate, such as a periodic forcing component where stochastic resonance is suspected. Even where the model can be reasonably

applied, there are several data pre-processing issues that must be tackled. First, if the record is unevenly spaced, it is reasonable to interpolate it, because the method deals with the kernel distribution, and filling data gaps would not affect the results of the analysis. The interpolation step should be chosen following the properties of the particular set. Second, some datasets require detrending. Since the method is generally applicable to quasi-stationary data, it makes sense to remove the obvious low-frequency trends, or to interpret the result of analysis of the raw data taking trends into account. For the derivation of the number of states, removing a general trend is helpful, because the technique operates a kernel distribution and provides more accurate results on data with some minimal detrending (like removing a moving average with appropriate window size). For the estimation of the noise level, the more accurate wavelet denoising is advisable.

6 Conclusion

The method of potential analysis that we propose is designed to identify the number of underlying states from climate time series data, to detect changes in the number of states when they occur, and to help locate associated bifurcations. We have shown some success in applying the method to known transitions in paleoclimate data, particularly the onset of Northern Hemisphere glacial cycles around 3 Myr ago. We have also identified a bifurcation in Greenland climate—the loss of the warm interstadial state—on approaching the last glacial maximum (Livina et al. 2010) and a degenerated potential for Marine Isotope Stadium 3 (Kwasniok and Lohmann 2009).

On applying the method to Holocene data, the predominant picture is of a single, stable global climate state. Only for particular time periods as e.g. 3.5 kyr BP, multiple states are detected over a time window of up to 1000 years. We find furthermore a single, stable global climate state at present, which appears to be undergoing a forced movement of this stable state due to anthropogenic forcing. There is no sign of global bifurcation and hence the global climate is not in a state of “runaway”. However, there are signs of the appearance of alternative climate states in particular regions at particular times during the Holocene. Hence the observational record should be examined more closely at a regional scale, and this is the focus of ongoing work.

A climatic record allows us to analyse the states of the system that have been observed and not those that may appear in future or existed prior to the record time scale. There may exist equilibrium states not experienced by the system in the analysed time series (for instance, snowball

earth or runaway greenhouse). The proposed method allows us to interpret the system's variability and to analyse its local stability, but may be unable to screen the complete phase space. However, the dynamics of the potential may be used for forecasting appearing states that have not previously existed, and we plan to develop the method in this direction.

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